

Homework 1

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3. Let P_n denote the statement that, for any $y_1, \dots, y_n \in S$ and any $\alpha_1, \dots, \alpha_n \geq 0$ satisfying $\sum_{i=1}^n \alpha_i = 1$, we have $\sum_{i=1}^n \alpha_i y_i \in S$. The implication $P_n \Rightarrow P_2$ is immediate by keeping two $\alpha_i \alpha_j > 0$ and $\alpha_i + \alpha_j = 1$, and let the rest $\alpha_k = 0$, $k \in [n] \setminus \{i, j\}$. For $P_2 \Rightarrow P_n$, we proceed by induction on n . The base case $n = 2$ is exactly P_2 . Assume that P_{n-1} holds. We prove P_n . Let $y_1, \dots, y_n \in S$, and let $\alpha_1, \dots, \alpha_n \geq 0$ satisfy $\sum_{i=1}^n \alpha_i = 1$. Define $\gamma = \sum_{i=1}^{n-1} \alpha_i = 1 - \alpha_n$. If $\gamma = 0$, then by nonnegativity, $\alpha_1 = \dots = \alpha_{n-1} = 0$. Hence $\alpha_n = 1$, and therefore $\sum_{i=1}^n \alpha_i y_i = y_n \in S$. Now suppose $\gamma > 0$. Define $\bar{y} = \sum_{i=1}^{n-1} \frac{\alpha_i}{\gamma} y_i$. Since $\frac{\alpha_i}{\gamma} \geq 0$ and $\sum_{i=1}^{n-1} \frac{\alpha_i}{\gamma} = 1$, the induction hypothesis P_{n-1} gives $\bar{y} \in S$. Moreover, $\sum_{i=1}^n \alpha_i y_i = \gamma \bar{y} + \alpha_n y_n$. Since $\gamma + \alpha_n = 1$, $\gamma, \alpha_n \geq 0$, and $\bar{y}, y_n \in S$, the two-point property P_2 implies $\gamma \bar{y} + \alpha_n y_n \in S$. Thus P_n holds. Therefore, by induction, P_2 implies P_n for every finite $n \geq 2$, and hence the two conditions are equivalent.

□

4. Consider the ambient space $\mathbb{R}^n$. Let $C \subseteq \mathbb{R}^n$, and let $L := \{x_0 + tv : t \in \mathbb{R}\}$ be an arbitrary line, where $x_0, v \in \mathbb{R}^n$ and $v \neq 0$.

($\Rightarrow$) Suppose C is convex. For any $x_1, x_2 \in C$ and any $\theta \in [0, 1]$, we have $\theta x_1 + (1 - \theta)x_2 \in C$. Now take any $x_1, x_2 \in L$. Since both points lie on the same line L , there exist $t_1, t_2 \in \mathbb{R}$ such that $x_1 = x_0 + t_1 v$ and $x_2 = x_0 + t_2 v$. Hence $\theta x_1 + (1 - \theta)x_2 = x_0 + (\theta t_1 + (1 - \theta)t_2)v \in L$. Therefore, for any $x_1, x_2 \in C \cap L$, we have $\theta x_1 + (1 - \theta)x_2 \in C \cap L$. Thus $C \cap L$ is convex.

($\Leftarrow$) Suppose $C \cap L$ is convex for every line L . To show that C is convex, take any $x_1, x_2 \in C$ and any $\alpha \in [0, 1]$. If $x_1 = x_2$, then $\alpha x_1 + (1 - \alpha)x_2 = x_1 \in C$. Otherwise, choose the line $L := \{x_1 + t(x_2 - x_1) : t \in \mathbb{R}\}$, which contains both x_1 and x_2 . Hence $x_1, x_2 \in C \cap L$. Since $C \cap L$ is convex, $\alpha x_1 + (1 - \alpha)x_2 \in C \cap L \subseteq C$. Therefore C is convex.

□

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8. (1) Let $S = \{x \in \mathbb{R}^3 : x_1^2 + 2x_2^2 \leq 4x_3\}$. Define $f(x) = x_1^2 + 2x_2^2 - 4x_3$. Since f is continuous and $S = \{x : f(x) \leq 0\}$, the set S is closed. Hence $\text{cl}(S) = S$. A point is an interior point precisely when the inequality is strict, so $\text{int}(S) = \{x \in \mathbb{R}^3 : x_1^2 + 2x_2^2 < 4x_3\}$. Therefore, $\partial S = \text{cl}(S) \setminus \text{int}(S) = \{x \in \mathbb{R}^3 : x_1^2 + 2x_2^2 = 4x_3\}$.

(2) Let $S = \{x \in \mathbb{R}^2 : 1 \leq x_1 \leq 4, x_2 = 2\}$. This is a closed line segment in $\mathbb{R}^2$, so $\text{cl}(S) = S$. Since S lies entirely on the line $x_2 = 2$, no open ball in $\mathbb{R}^2$ can be contained in S . Thus $\text{int}(S) = \emptyset$. Hence $\partial S = \text{cl}(S) \setminus \text{int}(S) = S$.

(3) Let $S = \{x \in \mathbb{R}^3 : x_1 + x_2 = 3, x_1 + 2x_2 + x_3 \leq 5\}$. Both $\{x : x_1 + x_2 = 3\}$ and $\{x : x_1 + 2x_2 + x_3 \leq 5\}$ are closed, so their intersection S is closed. Therefore, $\text{cl}(S) = S$. Since S is contained in the plane $x_1 + x_2 = 3$, it contains no open ball in $\mathbb{R}^3$, and hence $\text{int}(S) = \emptyset$. Therefore, $\partial S = \text{cl}(S) \setminus \text{int}(S) = S$.